02_exploratory_data_analysis

March 29, 2026

1 Supply Chain Optimization - Notebook 02: Exploratory Data Analysis

Project: Supply Chain Cost & Operational Efficiency Analysis
Dataset: Supply Chain DataSet - Amir Motefaker (Kaggle)
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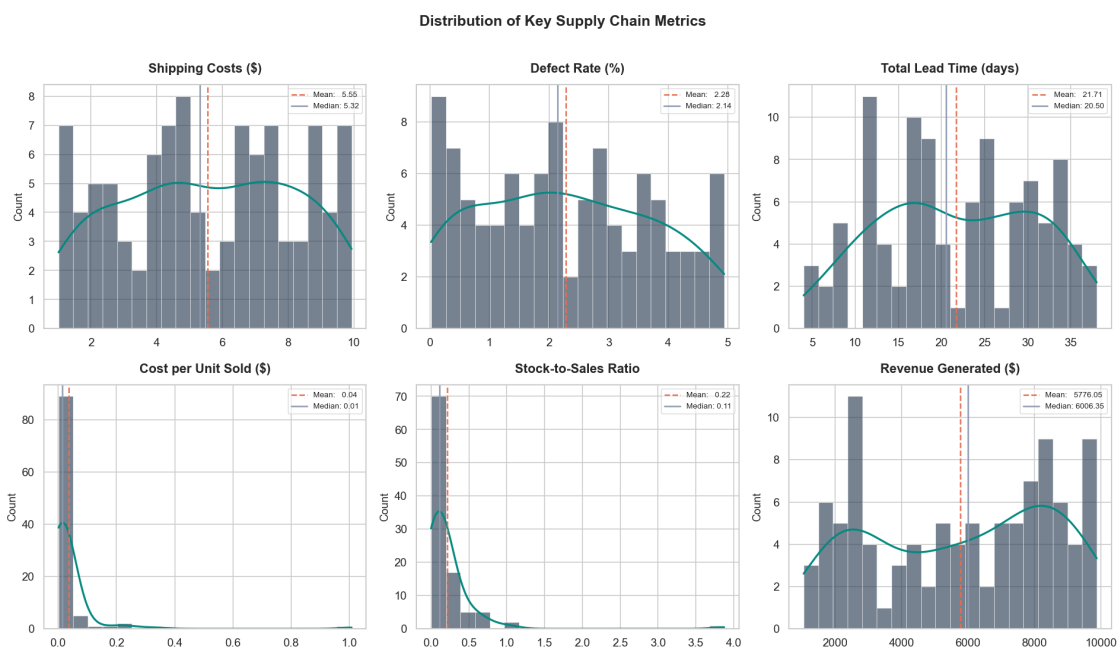
Date: *28/03/2026*

2 Exploratory Data Analysis

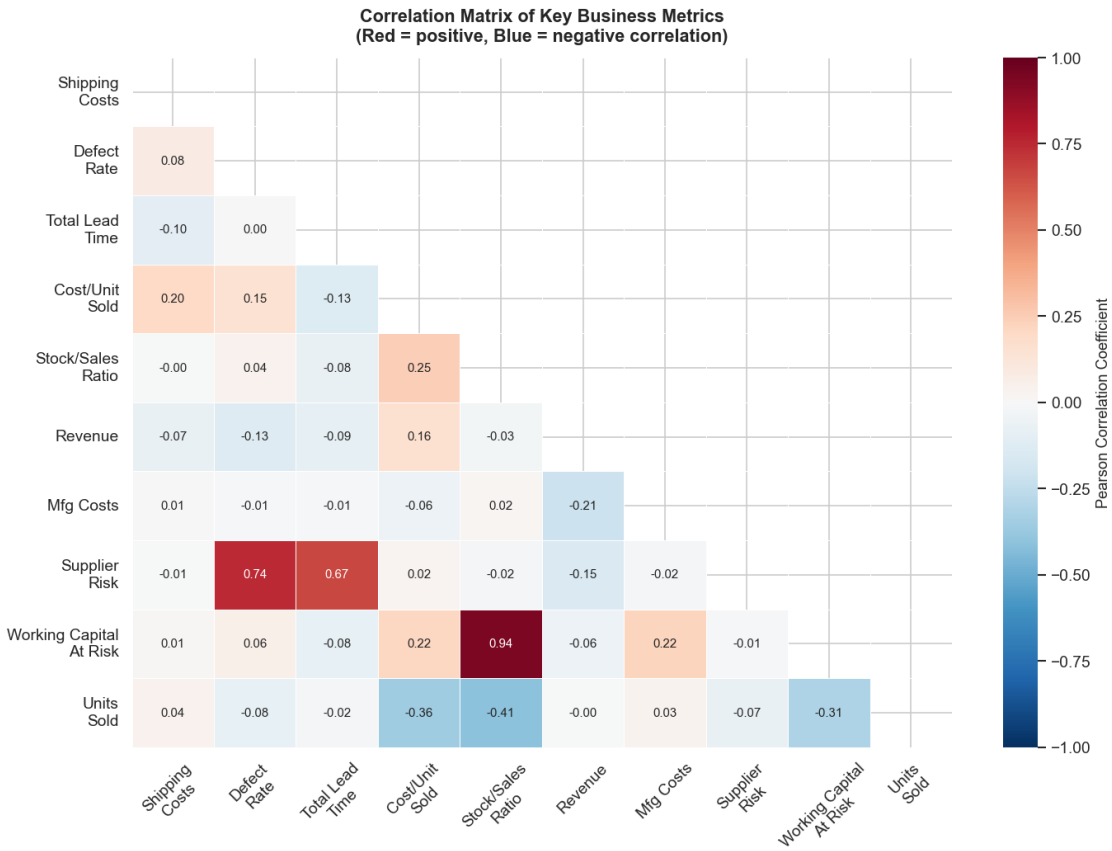
2.1 Distribution of Key Supply Chain Metrics

Let's take a look at some of the key variables in isolation first, to understand the shape and spread of distributions before combining variables.



Shipping costs, **total lead time**, **defect rates** and **revenue** are all approximately normally distributed.

Cost per unit sold and **stock-to-sales ratio** are both heavily right-skewed. A small number of SKUs are pulling the mean far above the median. For those two variables, medians will be used for all group comparisons to avoid outlier distortion.



- `Cost/Unit Sold` and `Units Sold` have a negative correlation of ~-0.36: SKUs with higher sales volume naturally have lower per-unit shipping cost, confirming the logic of our engineered metric.
- `Stock/Sales Ratio` and `Units Sold` are negatively correlated (~-0.41): slow-moving SKUs accumulate more stock relative to sales, as expected.
- `Defect Rate` and `Total Lead Time` show very weak correlation, meaning these variables are largely independent — suppliers that are slow are not necessarily more defective, and vice versa.

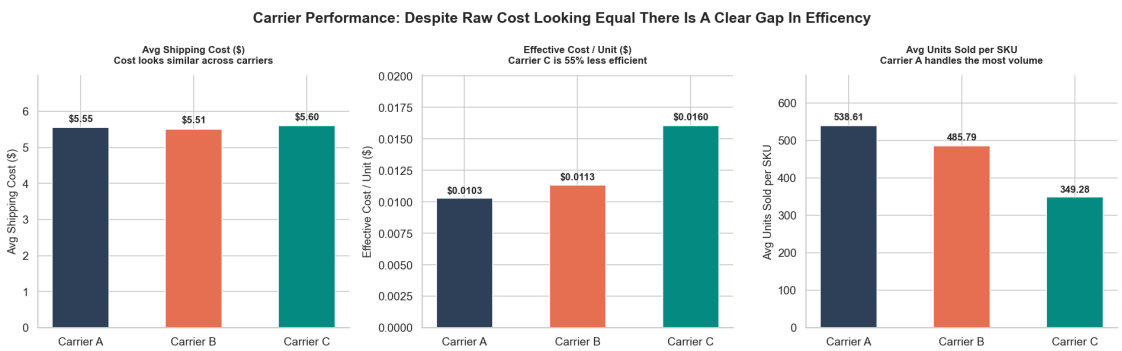
2.2 Shipping Cost Efficiency

Now I will focus on answering the question of shipping efficiency.

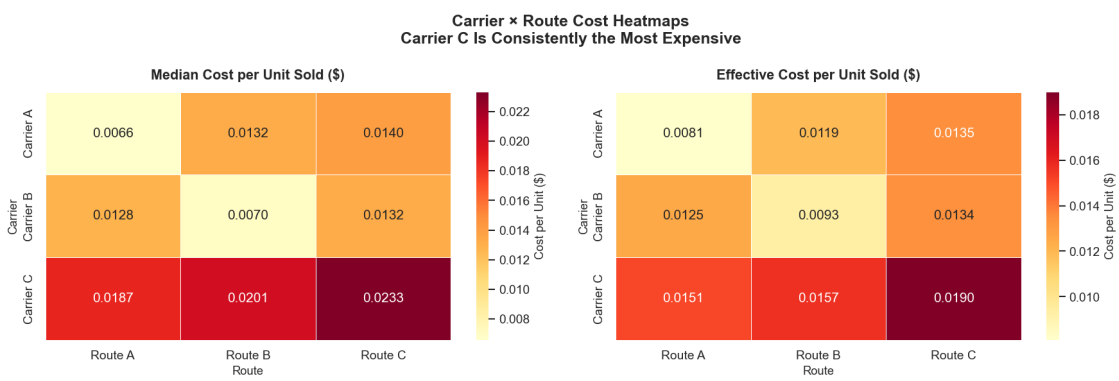
2.2.1 CARRIER PERFORMANCE TABLE

```
                  n_skus  avg_shipping_cost  median_cpu  effective_cpu  \
shipping_carriers
```

```
Carrier A              28              5.5549      0.0104          0.0103
Carrier B              43              5.5092      0.0104          0.0113
Carrier C              29              5.5993      0.0208          0.0160

                   avg_units_sold  efficiency_index
shipping_carriers
Carrier A               538.6071            1.0000
Carrier B               485.7907            1.1000
Carrier C               349.2759            1.5540
```



All three carriers charge near-identical average shipping costs (~$5.50). But Carrier C's effective cost per unit sold is **55% higher** than Carrier A's because Carrier C's SKUs sell far fewer units on average. This is the core inefficiency: the same shipping spend is spread over far fewer units with Carrier C, making each unit disproportionately expensive to ship.



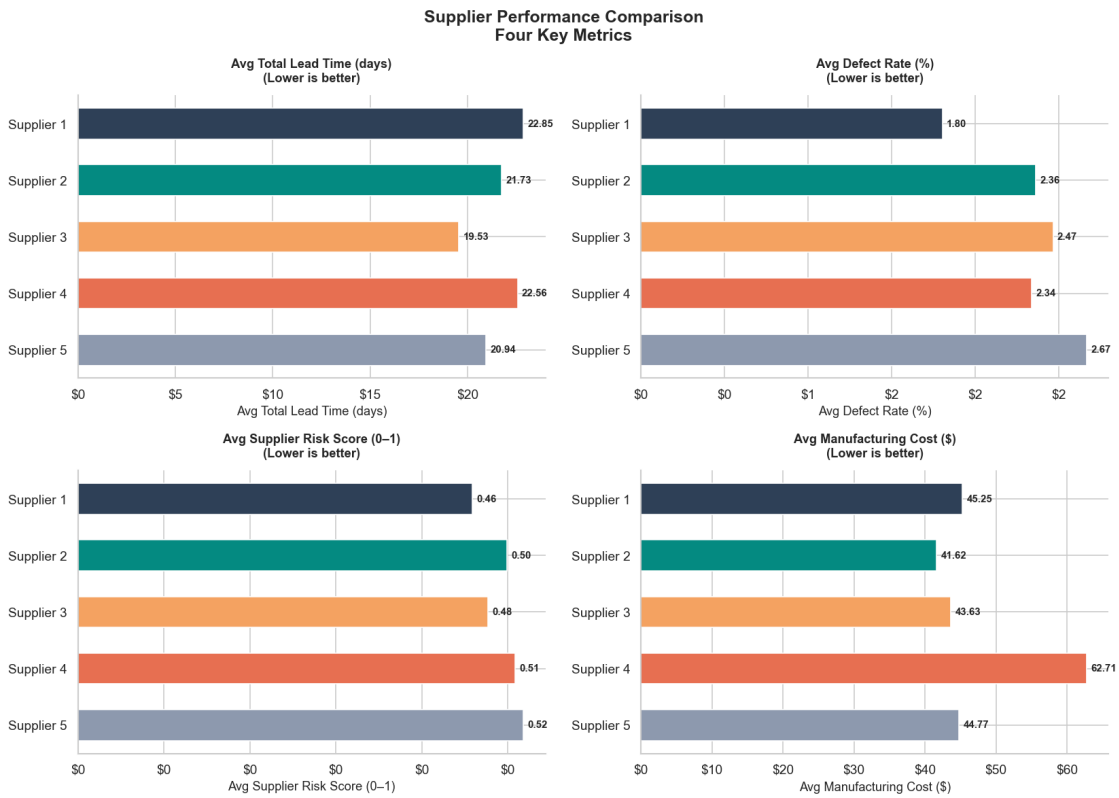
Both heatmaps tell a consistent story: **Carrier C is the most expensive operator across all three routes**, and this is not distorted by outliers (the median heatmap confirms it). Carrier A × Route A is the clear benchmark — the cheapest combination with the highest average volume.

2.3 Supplier Efficiency

Now, let's move on to the analysis of supplier efficiency.

2.3.1 SUPPLIER PERFORMANCE SCORECARD

```
               n_skus  avg_lead_time  avg_defect_rate  avg_risk_score  \
supplier_name
Supplier 1         27        22.8519           1.8036          0.4586
Supplier 2         22        21.7273           2.3628          0.4989
Supplier 3         15        19.5333           2.4658          0.4771
Supplier 4         18        22.5556           2.3374          0.5085
Supplier 5         18        20.9444           2.6654          0.5181

               avg_mfg_cost  fail_rate_pct  pass_rate_pct  revenue_per_cost
supplier_name
Supplier 1          45.2540        22.2000        48.1000          128.9300
Supplier 2          41.6225        36.4000        22.7000          137.0200
Supplier 3          43.6341        20.0000        13.3000          149.4200
Supplier 4          62.7097        66.7000         0.0000           76.6000
Supplier 5          44.7682        38.9000        16.7000          136.9300
```



Supplier 4 is the clearest underperformer in the dataset. It has the second highest average lead time, average defect rate, second highest composite risk score, 0% inspection pass rate and 66.7% fail rate and by far the highest manufacturing costs (`~$62.71 vs. a ~$43.82` average for the others). This combination of slower, lower quality, and more expensive makes Supplier 4 the primary target for renegotiation. Supplier 1, while having the highest lead time, makes up for it

by having the lowest defect rate by a large margin.

Supplier 3 stands out positively, it has the shortest lead time, defect rates within an acceptable range of average and competitive costs, making it the benchmark for what good supplier performance looks like in this dataset.

2.3.2 Financial Impact Quantification

```
#1 - MANUFACTURING COST PREMIUM

Supplier 4 avg mfg cost:          $62.71
Supplier 3 avg mfg cost (best):   $43.63
All other suppliers avg:          $43.88

Premium vs. Supplier 3:           $19.08/SKU
Premium vs. all others:           $18.83/SKU
Supplier 4 SKU count:             18

→ Estimated saving if matched Supplier 3: $343.36
→ Estimated saving vs. dataset avg:       $338.99

#2 - REVENUE DELAY OPPORTUNITY COST

Supplier 3 avg total lead time:  19.5 days (benchmark)
Supplier 4 avg total lead time:  22.6 days
Excess lead time vs. benchmark:  3.0 days per SKU

Avg revenue per SKU:             $5,776
Avg total lead time:             21.7 days
Implied daily revenue per SKU:   $266.05

→ Delay cost per Supplier 4 SKU: $804.08
→ Total delay cost (18 SKUs):    $14,473.38

#3 - DEFECT-RELATED MANUFACTURING WASTE

Supplier 4 avg defect rate:  2.337%
Other suppliers avg:         2.264%
Supplier 4 defect multiple:  1.03×

Supplier 4 est. waste:       $26.38
All others combined:         $81.45


CONSOLIDATED FINANCIAL IMPACT - SUPPLIER 4

  Mfg cost premium (vs avg):  $    338.99
  Revenue delay opp cost:     $  14473.38
  Defect-related waste:       $     26.38
```

```
  Total estimated impact:     $  14838.75
```

Revenue delay is the dominant component by far. Reducing Supplier 4's lead time to match Supplier 3's or at least the dataset average should be the priority.

2.4 Inventory Risk

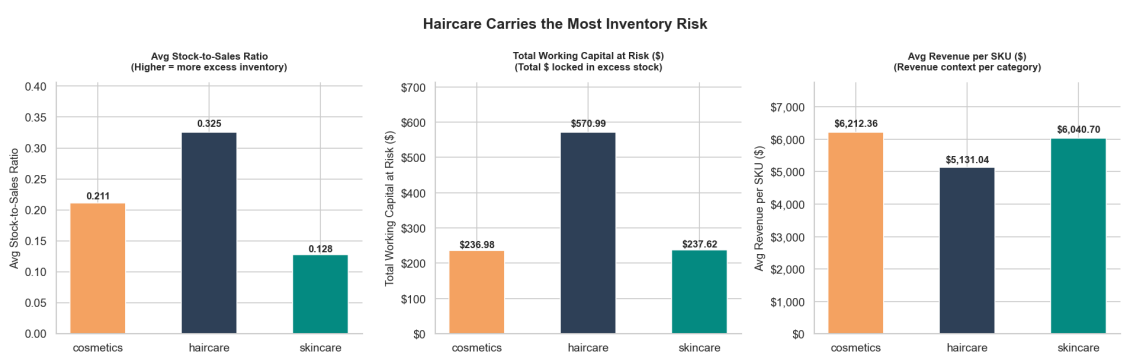
To finish off the exploratory data analysis section, I will look at the inventory to assess if there is any excess that poses financial risk.

2.4.1 INVENTORY LANDSCAPE SUMMARY

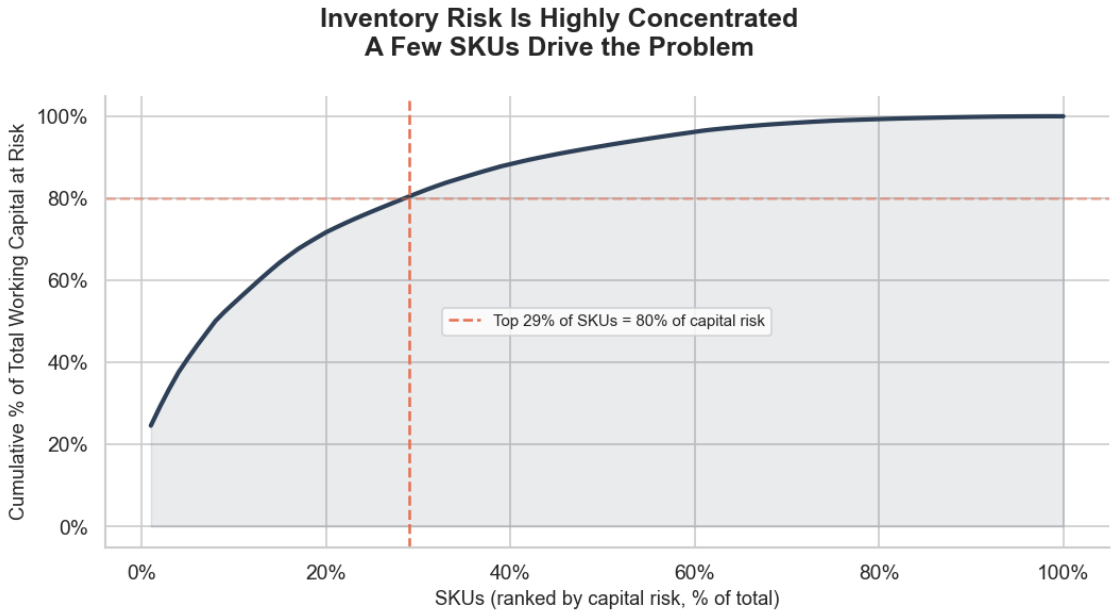
```
                              Mean   Median  Std Dev    Min      Max
Stock Levels (units)        47.7700  47.5000  31.3690 0.0000 100.0000
Units Sold                 460.9900 392.5000 303.7800 8.0000 996.0000
Stock-to-Sales Ratio (SSR)   0.2170   0.1150   0.4220 0.0000   3.8750
Working Capital at Risk ($) 10.4560   3.9900  26.9530 0.0000 256.9600


SKUs with SSR > 1.0 (more stock than units sold): 2
SKUs with SSR > 0.5 (moderate overstock):         11
Total working capital at risk across all SKUs:    $1045.5900
```

2.4.2 CATEGORY INVENTORY SUMMARY

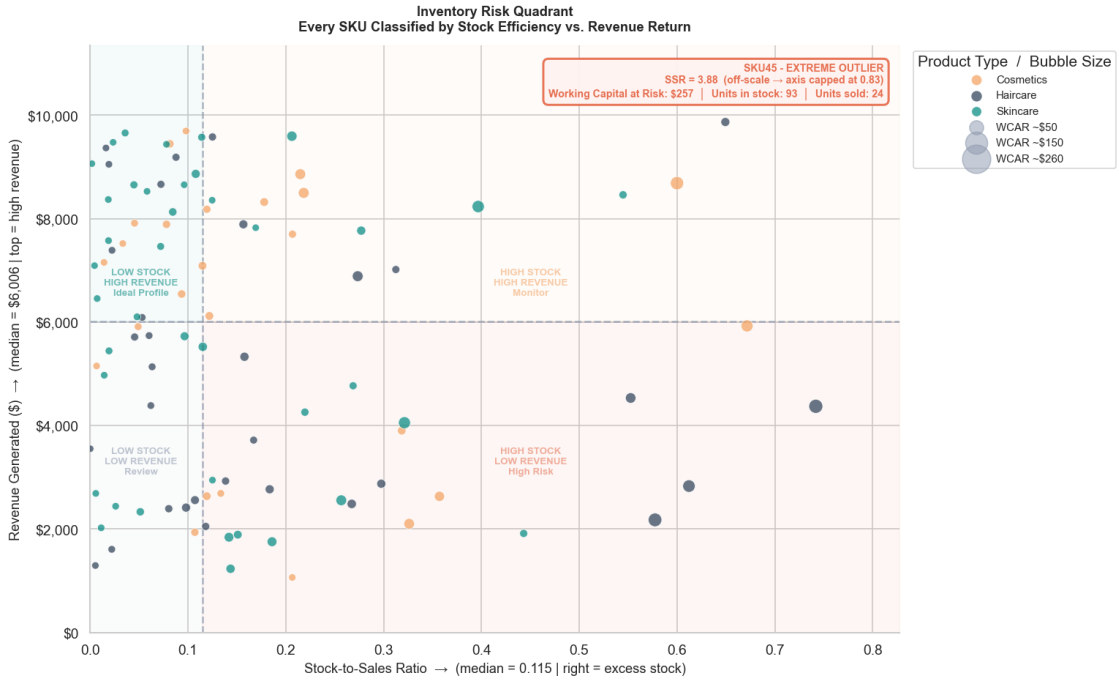
```
              n_skus  overall_ssr  avg_ssr  med_ssr  total_wcar  avg_wcar  \
product_type
cosmetics         26       0.1300   0.2110   0.1210    236.9800    9.1150
haircare          34       0.1210   0.3250   0.1220    570.9900   16.7940
skincare          40       0.0780   0.1280   0.0960    237.6200    5.9400

              wcar_share_pct
product_type
cosmetics            22.7000
haircare             54.6000
skincare             22.7000
```



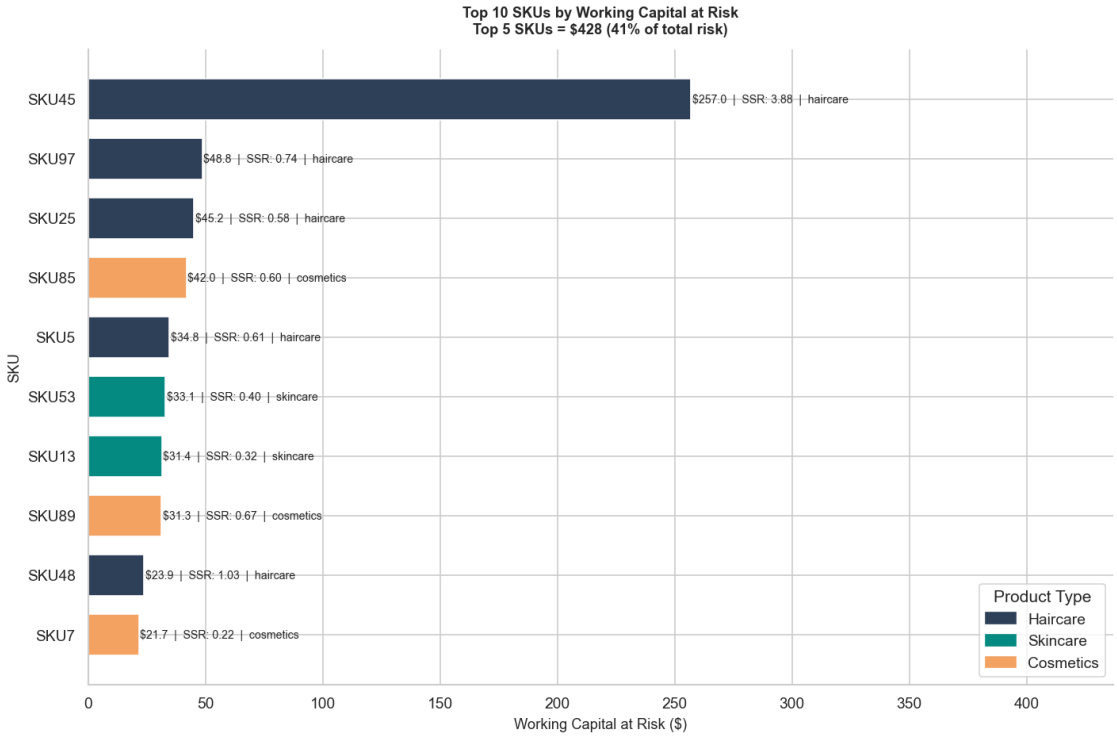
Haircare holds the most working capital at risk (~$571 total, 54% of all categories) and has the highest average SSR. Critically, haircare also has the lowest average revenue per SKU (~$5,131), meaning it generates the least return while tying up the most capital.



The cumulative capital risk curve has a steep initial slope, meaning that a small number of SKUs account for a disproportionate share of total capital risk. The implication is that targeted action on the top 5–10 SKUs will recover the majority of trapped capital, rather than requiring a broad portfolio-wide intervention.



30 SKUs sit in the 'High Stock / Low Revenue' danger zone — these are the highest-priority reduction targets. Notably, the largest bubbles (highest capital risk) are overwhelmingly haircare SKUs clustered in this quadrant. The 'Low Stock / High Revenue' quadrant in the top-left is the ideal profile, dominated by skincare — confirming that skincare's inventory management is the internal benchmark to replicate.



SKU45 accounts for a disproportionate share of total capital risk — its working capital at risk of `$256.96` is more than 5× the next-highest SKU (SKU97 at `$48.79`). The top 5 SKUs together account for the majority of total portfolio capital risk. A focused intervention on these 5 SKUs will unlock significantly more capital than a broad portfolio-wide stock reduction effort.

That concludes this notebook and part of the analysis. The final findings, recommendations and summary can be found in `Notebook 03: Summary`